LITERATURE REVIEW

Spillovers of US Interest Rates: Monetary Policy & Information Effects

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ANALYTICAL ABSTRACT

The paper asks why recent studies find that a US monetary tightening is followed by expansion abroad, contrary to the classical view of recession, currency depreciation and tighter financial conditions. Camara argues that the answer lies in the Fed Information Effect. Following Bauer and Swanson, the high-frequency surprise around FOMC meetings is regressed on public macroeconomic and financial news released before the meeting; the residual is labelled a pure monetary policy component (MP) and the fitted value an information component (FIE). Both components enter panel local projections for up to 50 countries over 1988 to 2019, and Bayesian structural vector autoregressions country by country. The MP component produces depreciation, higher long-term rates and falling output and equity prices. The FIE component produces the opposite, and the unadjusted surprise yields a blend of the two, which the author uses to explain the atypical findings. Results are reported to survive four alternative identification schemes and many sub-samples. The review finds the decomposition informative but the interpretation of the fitted component as an information shock only partly supported: it is a linear function of lagged global conditions, estimated in sample, and used as a regressor without correcting inference.

1 Introduction

1.1 Paper identification and classification

The paper is Camara, S., “Spillovers of US Interest Rates: Monetary Policy & Information Effects”, arXiv:2108.01026, in the economics general-economics category (econ.GN), version 2, revised in May 2024. The first draft is dated August 2021. The text carries no journal reference, so it is treated as a working paper. The paper address is arxiv.org/abs/2108.01026. The version read in full is the HTML rendering of version 2, including footnotes, equations, the bibliography and the textual description of the appendices. The figures themselves, which are impulse response plots, were not machine readable; every statement about them below relies on the author’s written description and is marked as such.

The paper is empirical and econometric. The field is international macroeconomics and the subfield is the international transmission of monetary policy, with a methodological component in the identification of monetary policy shocks through high-frequency data. It sits in the International Finance series because its outcome variables are exchange rates, long-term interest rates, equity prices and trade across fifty economies.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

Whether the sign of the international effect of a US interest rate increase depends on the reason the rate rises. A rise that reflects a policy choice made without reference to news should transmit differently from a rise that reflects the Federal Reserve responding to, or revealing, good news about the US economy.

WHY DOES THE QUESTION MATTER?

The US dollar is the dominant reserve, invoicing and funding currency, so Federal Reserve decisions are a first-order external shock for both advanced and emerging economies. A literature using high-frequency identification (Gertler and Karadi 2015; Ilzetzki and Jin 2021) reports that a US tightening is followed by foreign expansion. If correct, this overturns the textbook transmission mechanism; if it is an artefact of a contaminated shock series, policy conclusions about spillovers and the global financial cycle (Rey 2013; Miranda-Agrippino and Rey 2020) are biased.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Before the paper, classical evidence from vector autoregressions (Eichenbaum and Evans 1995; Dedola, Rivolta and Stracca 2017) supported the conventional view, while newer high-frequency evidence supported an atypical view. Separately, the US-only literature had shown that high-frequency surprises contain an information component (Romer and Romer 2000; Nakamura and Steinsson 2018; Jarocinski and Karadi 2020; Miranda-Agrippino and Ricco 2021; Bauer and Swanson 2023).

What was missing is a direct link: whether purging that component restores the conventional international result across a large panel of countries.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes that the standard surprise is a mixture of two shocks with opposite foreign effects. It decomposes the surprise by regression on pre-announcement news, estimates impulse responses by panel local projections (Jorda 2005) and Bayesian vector autoregressions, and finds that the MP component produces conventional responses, the FIE component produces opposite responses, and the standard surprise produces an average of the two.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The paper asks whether, in a panel of up to 50 economies between 1988 and 2019, the responses of the nominal exchange rate against the dollar, the long-term interest rate, consumer prices, industrial production and equity prices to a US monetary policy surprise differ in sign according to whether the surprise is the component orthogonal to, or the component predicted by, public news released before the FOMC announcement, and whether this difference accounts for the atypical positive foreign responses found with the unadjusted surprise.

2.2 Hypotheses

- H1. The pure monetary policy component (MP) causes foreign currency depreciation, higher long-term rates, falling industrial production and falling equity prices.
- H2. The Fed Information Effect component (FIE) causes the opposite: appreciation, no immediate rise in long-term rates, rising output and rising equity prices.
- H3. The Standard HFI response, obtained with the unadjusted surprise, is an average of the MP and FIE responses, which explains the atypical dynamics in earlier work.
- H4. These patterns are not driven by sample composition, time period, exchange rate regime, openness, or the particular method used to separate the two components.

2.3 Existing literature and research gap

PAPER The author organises the literature in three strands: studies of conventional negative spillovers (Eichenbaum and Evans 1995; Uribe and Yue 2006; Dedola, Rivolta and Stracca 2017); studies reporting atypical expansion (Canova 2005; Stavrakeva and Tang 2019; Ilzetzki and Jin 2021); and studies that already account for information effects when measuring spillovers (Jarocinski 2022; Degasperi, Hong and Ricco 2020). The stated gap is that the second strand does not control for the

information content of FOMC announcements, and the third strand does not report the spillovers of the information component itself.

CRITIQUE The gap is real but narrow. The decomposition is borrowed from US-focused work; the novelty is the application to a large international panel and the explicit reading of the FIE response. The paper also leaves open which interpretation of the fitted component is correct, a question that matters for the sign of the estimated effects (Section 7).

2.4 Claimed contribution

CRITIQUE The author claims three contributions: an identification scheme that purges information content and recovers conventional effects in the 2000s and 2010s; evidence that purging is necessary; and evidence that real and financial spillovers may be growing over time. The first two are supported qualitatively by the panel and by the country counts. The third rests on larger industrial production troughs in the 1998 and 2008 samples (about 12 percent and 20 percent against below 10 percent in the full sample, as described in the text); these are point estimates from different samples with different composition, no formal test of equality across samples is reported in the text, and the shock size is not stated, so the trend claim is suggestive only.

3 Theoretical Framework and Mechanism

The paper contains no structural model. Its mechanism is a reduced-form decomposition combined with an economic narrative. This section reconstructs the econometric objects in full and then adds the derivations needed to understand what they imply.

3.1 Decomposition of the monetary policy surprise

PAPER Following Bauer and Swanson (2023), the surprise is the first principal component of changes in the first four quarterly Eurodollar futures contracts in a window around each FOMC announcement. The surprise is regressed on a vector of news variables known before the announcement:

$$mps_t = \alpha + \beta' X_t + u_t, \tag{1}$$

- mps_t high-frequency monetary policy surprise attached to period t.
- X_t vector of six predictors: the non-farm payrolls surprise relative to the Money Market Services median; employment growth over the previous year; the log change in the S&P 500 over the previous 65 trading days; the change in the slope of the Treasury yield curve over three months (second principal component of 1 to 10 year zero-coupon yields); the log change in the Bloomberg commodity spot index over three months; and the implied skewness of the 10 year Treasury yield from options.
- α, β intercept and slope vector, estimated by ordinary least squares over the whole sample, February 1988 to December 2019.

u_t regression residual.

The two components are defined as

$$i_t^{MP} = mps_t - \hat{\alpha} - \hat{\beta}' X_t = \hat{u}_t, \tag{2}$$

$$i_t^{FIE} = \hat{\alpha} + \hat{\beta}' X_t. \tag{3}$$

Adding Equations (2) and (3) gives $mps_t = i_t^{MP} + i_t^{FIE}$ exactly. The economic idea is that a rational market prices the Fed reaction function imperfectly. The part of the surprise that moves with public news is the Fed responding to that news more strongly than markets expected; the remainder is a choice that does not respond to news.

INFERENCE Two algebraic properties follow from the least squares normal equations and are used throughout this review. First, the sample covariance between i_t^{FIE} and i_t^{MP} is exactly zero, because fitted values and residuals of a regression with an intercept are orthogonal. Second, therefore

$$\mathrm{Var}(mps_t) = \mathrm{Var}(i_t^{FIE}) + \mathrm{Var}(i_t^{MP}). \tag{4}$$

Define $w = \mathrm{Var}(i_t^{MP})/\mathrm{Var}(mps_t)$, the share of surprise variance that is the MP component. Then $1 - w$ equals the R^2 of Equation (1). The paper does not report this R^2 in the text read, which matters for interpreting the Standard HFI result (Section 8).

3.2 Local projections

PAPER Impulse responses are estimated by the panel local projection

$$y_{i,t+h} = \beta_h^{MP} i_t^{MP} + \beta_h^{FIE} i_t^{FIE} + \sum_{j=1}^{J_y} \delta_i^j y_{i,t-j} + \sum_{j=1}^{J_x} \alpha_i^j x_{i,t-j} + \sum_{j=1}^{J_i} \left(\phi_i^j i_{t-j}^{MP} + \varphi_i^j i_{t-j}^{FIE} \right) + \square_{i,t}. \tag{5}$$

$y_{i,t+h}$ outcome of country i, h months after t: nominal exchange rate against the dollar, long-term interest rate, consumer price index, industrial production, equity index.

$\beta_h^{MP}, \beta_h^{FIE}$ impulse responses at horizon h to each component; the object of interest, common across countries.

$x_{i,t-j}$ the other country-specific variables of the benchmark set.

J_y, J_x, J_i lag lengths: 2, 2 and 4, chosen by BIC and Hannan-Quinn for the first two and by efficiency arguments for the third.

$\square_{i,t}$ error; standard errors are clustered by month.

The coefficient β_h^{MP} is the effect, h months ahead, of a one-unit increase in the MP component holding fixed the FIE component and the lags. Because the two regressors are orthogonal by construction, each coefficient can be read as a separate causal object provided each component is exogenous to the outcome conditional on controls. The Standard HFI regression replaces the two components by mps_t and its lags.

INFERENCE A derivation shows why the Standard HFI coefficient is an average. Ignore lags and controls. If $y_{t+h} = \beta^{MP} u_t + \beta^{FIE} f_t + e_t$ with $f_t = i_t^{FIE}$ and $\mathrm{Cov}(u_t, f_t) = 0$, then a regression of y_{t+h} on $mps_t = u_t + f_t$ has probability limit

$$\beta_h^{HFI} = \frac{\mathrm{Cov}(y_{t+h}, mps_t)}{\mathrm{Var}(mps_t)} = \frac{\beta_h^{MP}\,\mathrm{Var}(u_t) + \beta_h^{FIE}\,\mathrm{Var}(f_t)}{\mathrm{Var}(u_t) + \mathrm{Var}(f_t)} = w\,\beta_h^{MP} + (1-w)\,\beta_h^{FIE}. \tag{6}$$

Hence the contemporaneous Standard HFI coefficient is a variance-weighted average of the two effects, with weights w and $1 - w$. This confirms the author's observation that the third column of the impulse response figures is an average, and it adds a precise statement: the average is exact only without lags and when the two effects do not vary across observations. With the lag structure of Equation (5) the equivalence is approximate. Equation (6) also implies that the Standard HFI coefficient changes sign when $w\beta_h^{MP} = -(1-w)\beta_h^{FIE}$, that is at $w_0 = \beta_h^{FIE}/(\beta_h^{FIE} - \beta_h^{MP})$. With opposite signs and equal absolute values $w_0 = 1/2$.

3.3 Alternative specifications

PAPER Robustness equations add a quadratic time trend and then country fixed effects:

$$y_{i,t+h} = \beta_h^{MP} i_t^{MP} + \beta_h^{FIE} i_t^{FIE} + \cdots + \mu_1 t + \mu_2 t^2 + \square_{i,t}, \tag{7}$$

$$y_{i,t+h} = \beta_h^{MP} i_t^{MP} + \beta_h^{FIE} i_t^{FIE} + \cdots + \mu_1 t + \mu_2 + \phi_i t^2 + \Gamma_i + \square_{i,t}, \tag{8}$$

and a version with US controls,

$$y_{i,t+h} = \beta_h^{MP} i_t^{MP} + \beta_h^{FIE} i_t^{FIE} + \cdots + \sum_j \alpha_j^{US} x_{US,t-j} + \square_{i,t}. \tag{9}$$

where $\cdots$ stands for the lag terms of Equation (5) and Γ_i is a country fixed effect. CRITIQUE In the printed Equation (8) the symbol ϕ_i is reused for a country-specific quadratic trend although ϕ_i^j already denotes lag coefficients of i^{MP}, and the constant μ_2 appears where Equation (7) has a quadratic coefficient; the notation is inconsistent but does not alter the estimation idea. The author notes that fixed effects with lagged dependent variables carry Nickell (1981) bias and uses Equation (8) only because Ilzetzki and Jin (2021) did.

3.4 Bayesian structural vector autoregression

PAPER For each country the model is a vector autoregression with exogenous regressor,

$$y_t = A_1 y_{t-1} + \cdots + A_p y_{t-p} + C x_t + \square_t, \qquad \square_t \sim \mathcal{N}(0, \Sigma), \tag{10}$$

with $n = 11$ endogenous variables (the five benchmark variables plus six US variables: two year Treasury rate, real trade-weighted dollar index, PCE price index, US exports, US imports, excess bond premium of Gilchrist and Zakrajsek 2012), $p = 12$ lags, and $m = 1$ exogenous variable, which is one of

the MP component, the FIE component or the Standard HFI surprise. A Minnesota prior (Litterman 1986) is imposed with autoregressive coefficient 0.8, overall tightness 0.1, cross-variable weight 0.5 and lag decay 1; the prior on the exogenous coefficient has mean 0 and variance 100. Estimation uses the ECB BEAR toolbox.

The pooled panel version stacks N countries and imposes common coefficients and a common innovation covariance:

$$Y_t = X_t\mathcal{B} + \mathcal{E}_t, \qquad \mathrm{vec}(Y) = (I_n \otimes X)\,\mathrm{vec}(\mathcal{B}) + \mathrm{vec}(\mathcal{E}), \qquad \bar{\Sigma} = \Sigma_c \otimes I_{NT}, \tag{11}$$

where Y_t is $N \times n$, X_t is $N \times (np + m)$ and $\mathcal{B}$ is $(np + m) \times n$. Errors are assumed uncorrelated across countries, $E(\square_{i,t}\square'_{j,t}) = 0$ for $i \neq j$. The author states that identification does not rely on Σ_c, because the shock enters as the exogenous regressor x_t.

CRITIQUE Equation (11) restricts all countries to the same dynamics and the same response coefficient, so the pooled model is far more restrictive than Equation (5), which allows country-specific lag coefficients. The zero cross-country error correlation is implausible when all countries respond to the same shock; the author acknowledges this about the older ordering approach (footnote 39) but the same assumption is retained in the pooled covariance. The mean-group model averages demeaned variables across countries and estimates one small system on the average, which is a different estimand again. The three SVAR designs therefore agree in sign but are not independent checks of the same coefficient.

3.5 Economic mechanism

PAPER The conventional mechanism: a tightening raises US yields and the dollar, depresses the world economy through trade, financial conditions and commodity prices, and forces local rates up. The information mechanism: when the Fed raises rates in response to strong data, or signals that the economy is stronger than expected, private agents revise growth expectations upward; foreign output, equity prices and currencies respond to the better outlook and to commodity demand, and the rate rise is a symptom rather than a cause.

CRITIQUE The paper does not test the mechanism. It does not estimate effects on expectations, risk premia, commodity prices as an intermediate step, or trade volumes by partner. It documents that the sign of the response differs, then labels the cause as information. This is discussed in Section 7.

4 Data and Empirical Strategy

4.1 Data

PAPER The sample includes every country with data for five variables from IMF International Financial Statistics or the OECD between January 1988 and December 2019: the nominal exchange rate against the dollar, a long-term interest rate, a consumer price index, an industrial production index and an equity index (in local currency, per footnote 16). The result is an unbalanced monthly panel of 50 countries, 28 advanced and 22 emerging economies (IMF 2019 classification). For countries without a long-term rate, it is constructed as the JP Morgan EMBI spread plus the short-term US risk-free rate from Kenneth French's data library, following the author's earlier work. Robustness samples replace the nominal rate by the trade-weighted real exchange rate (Darvas 2012; normalised so that an increase means depreciation), and merge euro area members into one unit. Monetary surprises begin in February 1988.

Table 1. **Data and sample as described in the paper**

Item	Description	Comment
Sample period	January 1988 to December 2019, monthly	Surprises start February 1988
Countries	50 (28 advanced, 22 emerging)	Unbalanced panel; 40 in country-by-country SVAR
Outcome variables	Exchange rate, long rate, CPI, industrial production, equity index	Domestic currency equity index
Shock series	First principal component of four Eurodollar futures changes	Bauer and Swanson (2023)
Predictors in first stage	Payroll surprise, employment growth, S&P 500, yield slope, commodities, Treasury skewness	Six variables
Lags	2 on outcomes and controls, 4 on shock components	BIC and Hannan-Quinn
Standard errors	Clustered by month	Common shock across countries

Source: reviewer's summary of Section 2 and footnotes 10 to 14 of the paper (official content of the paper; table layout generated by the reviewer). The number of observations is not stated in the text read.

Table 2. **Alternative schemes that separate the two components**

Scheme	How the pure shock is defined	Sample used by the author
Bauer and Swanson (2023), benchmark	Residual of surprise on public news	February 1988 to December 2019
Bu, Rogers and Wu (2021)	Fama-MacBeth two-step shock, largely unpredictable	January 1994 to September 2019
Jarocinski and Karadi (2020)	Sign restrictions on co-movement of rates and equities	January 1990 to December 2016
Miranda-Agrippino and Ricco (2021)	Residual of surprise on own lags and Greenbook forecasts	January 1991 to December 2015
Miranda-Agrippino and Nenova (2022)	Rates and S&P 500 co-movement applied to target, path and QE surprises of Swanson (2021)	Not stated in the text read
Nakamura and Steinsson (2018)	No purging; contrast case that reproduces atypical results	Not stated in the text read

Source: reviewer's summary of Section 4.1 of the paper (official content of the paper; table layout generated by the reviewer).

4.2 Empirical specification and identification

PAPER The identifying variation is the month-to-month variation in the high-frequency surprise around FOMC meetings, split by Equation (1). Countries are hit by the same shock simultaneously, so the useful variation is in time; this is why standard errors are clustered by month. The panel dimension pools many outcome series to estimate a common response coefficient, while country-specific lags absorb country dynamics.

CRITIQUE The effective number of independent shock observations is the number of FOMC-related months, not countries times months. Exogeneity of each component is assumed on three grounds: the narrow window around the announcement, the orthogonality of the MP residual to selected news, and the assertion that foreign variables cannot affect the surprise. The last point is delicate for the FIE component, as discussed below. The paper does not state in the text read how surprises are aggregated to months with no meeting or multiple meetings, nor the units or size of the shock; hence magnitudes are not interpretable in intuitive quantities, such as per standard deviation of the shock.

4.3 Data limitations and biases

- Look-ahead in the decomposition: $\hat{\alpha}$ and $\hat{\beta}$ in Equations (2) and (3) are estimated on the whole 1988 to 2019 sample, so the components are not realised-time quantities, and the FIE component embeds future information about the sample.
- Generated regressors: i_t^{MP} and i_t^{FIE} are estimated and then used as regressors; the text reports clustering by month but no correction for first-stage estimation error (Pagan 1984).
- Unbalanced panel and composition: countries enter and leave; the author checks by dropping those that leave and re-enter. Emerging economies have shorter and noisier series, which affects the pooled coefficient.

- Constructed long-term rates for countries without an OECD series mix an EMBI spread and a US short rate, which is a different object from a domestic government bond yield and may move mechanically with US rates.
- Cross-country comparability: harmonised IMF and OECD sources limit definitional heterogeneity, but equity indices and industrial production differ in coverage, and mining, construction and retail indices have thinner coverage (acknowledged by the author).
- Data mining: roughly 180 reported figures and several identification schemes allow many comparisons; the paper reports qualitative sign agreement and does not adjust for multiple comparisons.

5 Results

EVIDENCE All results are described through impulse response figures. Since the figures are not machine readable here, numbers below are those quoted in the text. Confidence bands are 68 and 90 percent, as stated by the author.

5.1 Benchmark: opposite responses to the two components

For the MP component, the nominal exchange rate depreciates on impact and further in the first year, with significance at 90 percent for the first four months. Long-term rates rise by 100 basis points on impact and peak near 200 basis points two months later, significant at 90 percent up to a year. Consumer prices rise slowly and become distinguishable from zero only at 68 percent after a year. Industrial production falls, with a trough 7 to 8 months out, and equity indices fall about 10 percent initially and reach a trough near 25 percent, recovering after about a year.

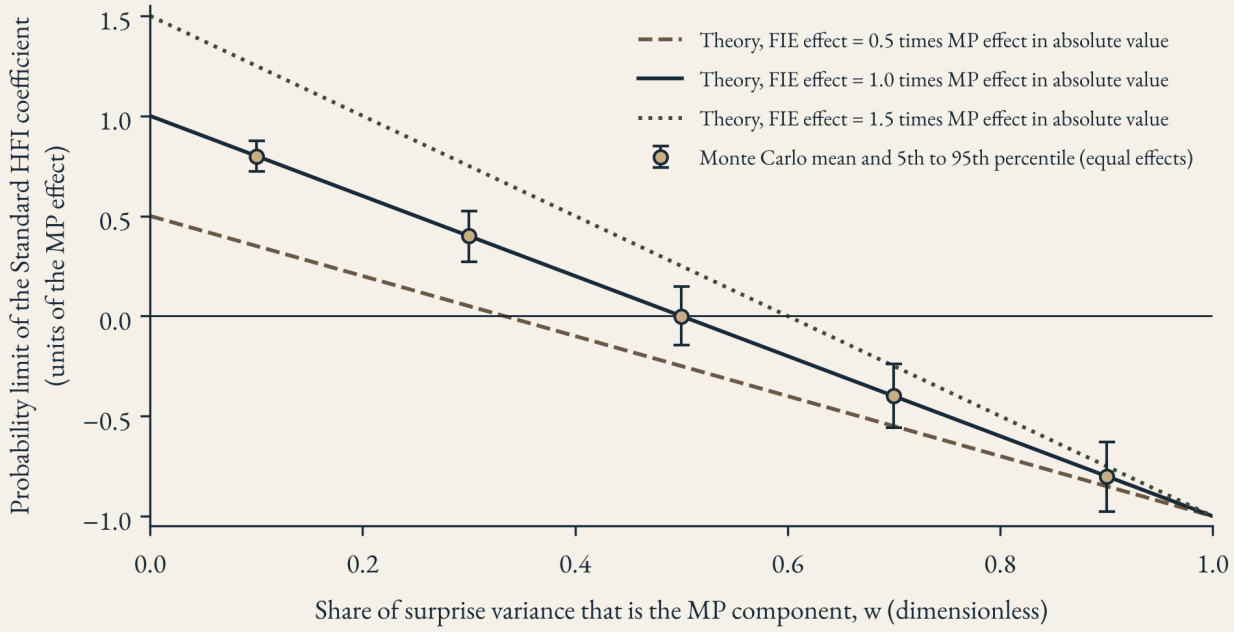
For the FIE component, the exchange rate appreciates persistently (hump-shaped, significant at 68 percent for up to 18 months); long rates do not react on impact and rise only after 6 to 9 months; consumer prices rise slightly then fall below baseline after 9 months; industrial production rises, significant at 90 percent at all horizons shown, peaking at 16 to 18 months; equity indices expand and stay above baseline for up to 24 months.

CRITIQUE The magnitudes cannot be translated into intuitive units because the shock normalisation is not given. Taken at face value, a 25 percent equity trough and a 200 basis point long-rate peak would be large for a monetary surprise in the range typical of the Eurodollar-based series; this suggests either a large normalised shock or mixed-economy aggregation, and the issue deserves checking against the replication files.

5.2 Standard HFI as an average

EVIDENCE The Standard HFI regression produces, for industrial production, a significant increase on impact persisting for more than 12 months, and for equities a short fall followed by a rise above baseline after nine months. The nominal exchange rate depreciation is not statistically different from zero. The author concludes that the standard approach underestimates the depreciation after a pure shock and produces the atypical expansion. This is consistent with Equation (6): the Standard HFI coefficient is $w\beta^{MP} + (1-w)\beta^{FIE}$.

Figure 1. Probability limit of the Standard HFI coefficient as a function of the MP share of surprise variance



Source: generated by the reviewer from Equation (6). Lines are the theoretical weighted average with the MP effect normalised to minus one. Points are Monte Carlo means from 2000 simulated samples of 380 observations each, one predictor, equal and opposite effects, with normal noise; error bars show the 5th and 95th percentiles. Illustration only; the values are not estimates from the paper.

Figure 1 makes the mechanism explicit. If the FIE effect is as large as the MP effect and the MP share is above one half, the Standard HFI coefficient has the sign of the MP effect, and below one half it has the opposite sign. The paper's finding that the Standard HFI industrial production response is positive therefore requires, within this algebra, that the FIE part is not small relative to the MP part, or that the FIE effect is larger in absolute value. Since $1-w$ is the R^2 of the first-stage regression, which is known to be modest in the US literature, the large FIE weight needed is notable; the paper does not report the R^2, so this requirement cannot be verified from the text.

5.3 Sub-samples

EVIDENCE For advanced and emerging economies separately the opposite-sign pattern holds. Differences reported: after the MP component emerging economies see persistently rising consumer prices while advanced economies see significant falls, which the author links to import dependence (studies cited by the author on import dependence); long-term rates rise by 250 basis points on impact in emerging economies, against a slower and smaller increase in advanced ones. For samples starting in 1998 and 2008, industrial production troughs are deeper, near 12 and 20 percent.

5.4 Additional variables and trade

EVIDENCE The MP component raises domestic policy and lending rates immediately and reduces construction, mining and retail indices, raises unemployment (hump-shaped), and lowers commodity export and import prices and trade values. The FIE component delays rate pass-through by 6 to 9 months, raises construction, mining and retail activity, lowers unemployment for up to 18 months and raises trade values and commodity prices. The Standard HFI shows policy rates rising for up to two years, activity and commodity prices rising, and unemployment falling.

Table 3. **Reported responses by component, as described in the text**

Variable	MP component	FIE component	Standard HFI
Nominal exchange rate	Depreciation, significant at 90% for four months	Appreciation, significant at 68% up to 18 months	Depreciation not significant
Long-term rate	+100 bp on impact, peak +200 bp at two months	No impact; rises after 6 to 9 months	Not described in detail
Industrial production	Falls; trough 7 to 8 months	Rises; peak 16 to 18 months	Rises, significant for over 12 months
Equity index	About minus 10% on impact, trough near minus 25%	Rises, above baseline for 24 months	Short fall, then rise after 9 months
Unemployment	Hump-shaped rise	Falls for up to 18 months	Falls

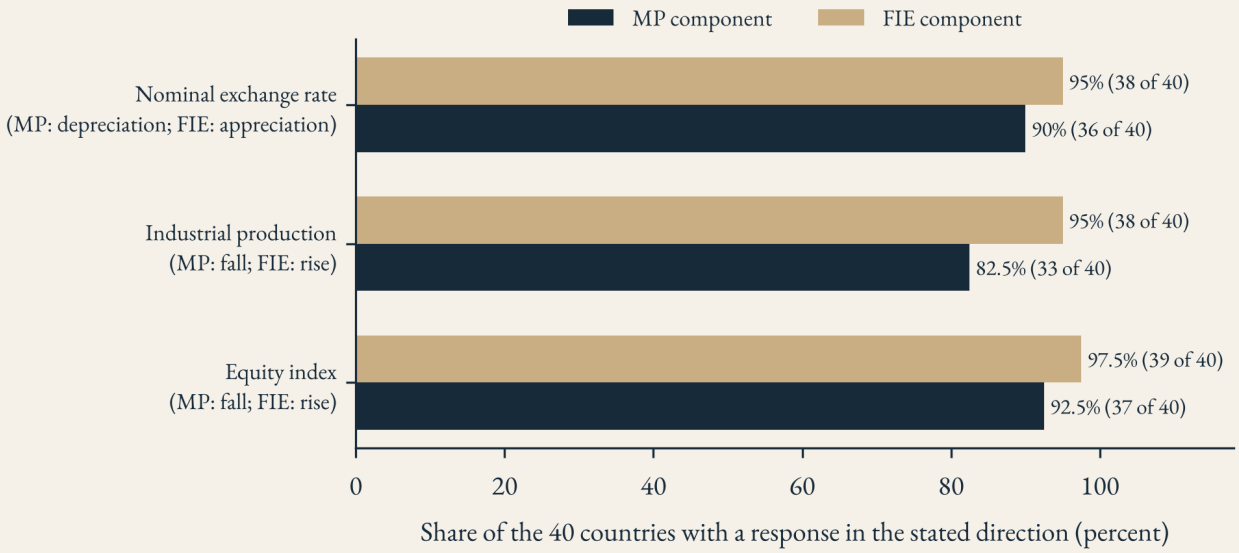
Source: Sections 3 and 4.1 of the paper; text description, official. Table layout generated by the reviewer. Shock normalisation is not stated in the text read.

Reading the table: the question answered is whether the sign of each response differs by component; the dependent variables differ by row; the specification is Equation (5) for the first two columns and its Standard HFI version for the third. The most important entry is industrial production, because it is the one on which the atypical literature rests. What the table cannot establish is the economic size of any entry, because shock size is missing, and it cannot show whether the MP and FIE responses are statistically different from each other, since the figures show each against zero.

5.5 Country-by-country Bayesian SVAR

EVIDENCE In 40 countries, the MP component generates depreciation in 90 percent, falling industrial production in 82.5 percent and falling equities in 92.5 percent. The FIE component generates appreciation in 95 percent, rising output in 95 percent and rising equities in 97.5 percent. The exceptions are listed in the paper (for example, China, Finland, India, Venezuela and Ukraine for the exchange rate response to MP).

Figure 2. **Share of countries showing the reported direction of response, country-by-country SVAR**



Source: counts quoted in Section 4.1 of the paper (official); percentages of 40 countries; chart generated by the reviewer. The direction is as reported for each component, not the same sign for both.

The heterogeneity pattern is reassuring about outliers: the opposite-sign result is not driven by a handful of economies. But the counts are sign counts at the author's significance convention, and the prior has a different relative weight in countries with shorter samples.

6 Robustness and Additional Tests

PAPER The author reports robustness along four lines: alternative identification schemes, alternative estimators, alternative samples and alternative variables. CRITIQUE Each line is assessed below against the specific threat it is meant to address, in the form concern, test, result and assessment.

Table 4. **Robustness tests: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Results depend on the particular decomposition	Four alternative schemes: Bu et al.; Jarocinski and Karadi; Miranda-Agrippino and Ricco; Miranda-Agrippino and Nenova	Qualitatively similar opposite-sign responses; industrial production muted for J-K full sample	Partly. Schemes share sources of contamination and different interpretations, so agreement shows robustness to method, not that the cause is information
Euro area drives the result	Euro area merged into one unit; real effective exchange rate replacing nominal	Same signs; real exchange rate depreciates after MP, appreciates after FIE	Yes for the euro concern; no for exchange rate regime effects beyond this
Equity fall driven by depreciation	Compare magnitudes in local currency	Equity drop 50 to 100 percent larger than depreciation	Yes, for the narrow concern
Advanced or emerging composition	Separate AE and EME panels	Same sign; larger long-rate and price effects in EMEs	Yes for sign; magnitudes differ
Time instability	Samples from 1998 and 2008	Same sign; deeper industrial production troughs	Partly. No test of equality across samples
Trends, fixed effects, Nickell bias	Equations (7) and (8)	Results unchanged	Partly. Fixed effects with lagged dependent variable biased; author notes this
Omitted US variables	Equation (9) with US controls	Effects slightly larger and more significant	Partly. The first-stage predictors are not all included
Estimation method	Country-by-country Bayesian SVAR; pooled and mean-group panel SVAR	Same signs; 90 to 97.5 percent of countries	Partly. Priors, long lags and pooling restrictions matter
Regime and openness heterogeneity	Exchange rate regime, current account openness, trade openness, GDP per capita, continents	Same signs; smaller exchange rate effect under pegs	Yes for sign; no formal heterogeneity tests
Sample choice of earlier literature	Ilzetzki and Jin dataset and specification	MP depreciation and fall; FIE appreciation and rise	Yes, strongest evidence on the reconciliation claim
Long rate construction	EMBI spread replacing long rate	MP raises spreads; FIE lowers them	Partly. Measures risk premium, not bond yield
Lag choice	12 lags on outcomes; zero lags on shocks	Reported in appendix	Not assessable from the text read

Source: Sections 3, 4.1 and 4.2 of the paper (official content); assessment column generated by the reviewer.

CRITIQUE The tests largely vary the sample or the estimator while leaving unchanged the construction of the shock components, which is where the central threat lies. The alternative identification schemes are the closest to a direct test; but they all aim at the same US-side contamination and could share the same limitation: a component correlated with the global cycle.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

The sign of the foreign response to the Bauer and Swanson decomposition differs between the two components across a very wide set of samples and estimators, and the unadjusted surprise yields a blend. The result is not driven by a few countries, by the euro area, or by the sample of Ilzetzki and Jin (2021).

SUPPORTED BUT CONDITIONAL

That the atypical dynamics of the earlier literature arise from the omission of the FIE component. This depends on the decomposition being valid, on Equation (6) holding approximately with the lag structure used, and on the first-stage R^2, which is not reported.

SUGGESTIVE

That the FIE effect operates through revised beliefs about the strength of the US economy, and that spillovers grow over time.

NOT ESTABLISHED

Causality of the FIE effect as an information channel; economic magnitude in intuitive units; statistical difference between the two responses; the comparison with equal-weighted alternatives; and a stable time trend of spillovers.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- A larger panel than comparable monthly studies (50 countries against 21 for Ilzetzki and Jin and 30 for Degasperi et al.), which makes composition effects less likely.
- Several identification schemes and two estimation families (local projections and Bayesian SVARs), reducing the risk that the result reflects a method.
- Reconciliation of both the conventional and the atypical findings within one framework.

STRENGTHS IDENTIFIED INDEPENDENTLY

- Replication on the Ilzetzki and Jin sample is the most informative test, because it uses the rival data and specification and still recovers the opposite-sign pattern.

- The averaging result (Equation 6) gives the contribution a clear and testable econometric form: the sign of the Standard HFI response depends on w.
- The country counts (36 of 40 depreciations, 38 of 40 appreciations after FIE) offer a distribution-free summary that does not depend on pooled coefficients.
- Transparent disclosure of Nickell bias, of the changed SVAR design in footnote 39, and of contrary evidence (Acosta 2022).

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The FIE component is the fitted value of a regression on six public variables that are themselves indicators of the US and global cycle (stock returns, commodity prices, payroll news, yield slope). It is therefore a smooth linear index of recent global conditions, not an information shock in the sense of a news event on the day of the meeting. **Why it matters.** The foreign response to such an index could simply reflect that a strong global cycle raises foreign output, equities and currencies, with the Fed raising rates because the same conditions prevail. The LP in Equation (5) controls only for lags of country variables and of the components, not for X_t itself, so the response to the FIE component is not cleanly separated from the response to the cycle. **Potential bias.** Positive bias in β_h^{FIE} for output and equities, and a spurious sign contrast with MP. **How it could be tested.** Add X_t and its lags as controls (a horse race of X_t against i_t^{FIE}); build a placebo with fitted values from a regression on predictors with no link to the Fed; replace the in-sample coefficients by rolling out-of-sample coefficients; and compare with the Jarocinski-Karadi shock, which uses the sign of the equity response on the day.

OTHER WEAKNESSES

- Generated regressors and look-ahead: the components are estimated in sample and used as regressors without correcting inference. The bias is typically in the standard errors and could either overstate or understate significance; a joint bootstrap of both stages would test it.
- Unreported first-stage fit: without the R^2 the weights w and $1-w$ in Equation (6) cannot be assessed, so the claim that the Standard HFI is an average cannot be turned into a numerical prediction and checked.
- No test of antisymmetry: the paper compares each response against zero but does not test $\beta_h^{MP} = -\beta_h^{FIE}$ or $\beta_h^{MP} \neq \beta_h^{FIE}$ with a covariance estimate. The visual contrast of two confidence bands is not a test of their difference.
- Interpretation of FIE is unsettled in the literature (signalling, disclosure of the Fed's information, or the Fed responding to news). The paper takes no position, yet the conclusion is phrased as a Fed Information Effect, whose bias implications differ across interpretations.
- Reliance on a single common shock series for 50 economies and about 380 months; adding countries increases the number of outcome series, not the number of independent shocks, so precision gains are limited.

- Bayesian SVAR with 11 variables and 12 lags relies heavily on the Minnesota prior in short country samples; the stated prior parameters are described as standard but results are not shown under alternatives in the text read.
- Exogeneity of the shock to foreign conditions is argued rather than tested; contemporaneous reaction of the Fed to foreign events cannot be excluded for the FIE component.
- Shock units unknown; there is no discussion of nonlinearities or asymmetry between tightening and easing, and no separation of the zero lower bound period.

8 Contradictions and Edge Cases

CRITIQUE The following uses direct reasoning and, where useful, illustrative calculation from Equation (6); numbers are generated by the reviewer and are not estimates from the paper.

When the Standard HFI atypical result should disappear. Suppose $\beta^{MP} = -1$ and $\beta^{FIE} = +\rho$ in units of the MP effect. The Standard HFI coefficient is $-w + \rho(1-w)$. With $\rho = 1$ and $w = 0.5$ it is zero; with $w = 0.7$ it is -0.4, the conventional sign; with $w = 0.3$ it is $+0.4$, the atypical sign. With $\rho = 1.5$ and $w = 0.5$ it is $+0.25$. Hence the atypical sign emerges only when the FIE effect is either as large as the MP effect and carries at least half the variance, or larger than the MP effect. If the first-stage R^2 were small, say 0.2, then $w = 0.8$ and even $\rho = 2$ gives $-0.8 + 0.4 = -0.4$, which would not produce an expansion. So the explanation of the atypical literature requires either a large predictable share or a much larger FIE effect, both of which are testable.

Sign flip at the other extreme. If $w \to 1$ (the surprise is almost entirely orthogonal to news), the decomposition is irrelevant and Standard HFI equals the MP response; if $w \to 0$ the surprise is almost entirely predictable and the Standard HFI response equals FIE. Real FOMC surprises sit in between, and the weight likely varies across sub-periods (for example during the zero lower bound or after 2008), so a constant w is an approximation; the deeper troughs in the 1998 and 2008 samples could simply reflect a changing w.

Mechanism failure. If monetary policy surprises carry a risk-premium component unrelated to growth news, the FIE component may capture changes in global risk appetite, which affect emerging market currencies and equities with the sign reported even in the absence of an information story. The mechanism would also fail if foreign central banks react to the Fed in ways correlated with X_t.

Dependence on pegs. The reported smaller exchange rate response for pegs is mechanical: the response of the nominal exchange rate to either component is limited by regime, so signs should be tested on interest rates and reserves for those countries.

Alternative mechanism generating the same pattern. A global demand shock raises US yields, commodity prices, equities and foreign activity jointly. Regressing the surprise on S&P 500 and commodity changes would assign that shock to FIE, which produces appreciation and expansion with no information channel at all.

Outside the original setting. The result is for 1988 to 2019, when the Federal Reserve communicated increasingly through forward guidance. It may not extend to a regime with rapid tightening and high inflation, where news is dominated by inflation surprises, which are not among the six predictors.

9 Further Literature

Table 5. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Kuttner (2001); Gurkaynak, Sack and Swanson (2005)	Futures-based measurement of policy surprises	Foundation of high-frequency identification	Source of the surprise series
Romer and Romer (2000)	The Fed reveals information about the economy	Preceded; first evidence of information in policy moves	Origin of the information-effect idea
Gertler and Karadi (2015)	High-frequency instruments in VARs	Basis of the standard approach that the paper challenges	Defines Standard HFI
Nakamura and Steinsson (2018)	Positive output response to a rate surprise via information	Alternative interpretation; the paper reproduces their atypical result	Source of the disclosure view
Jarocinski and Karadi (2020)	Sign restrictions separate pure and information shocks	Alternative identification used for robustness	Different interpretation, similar sign pattern
Miranda-Agrippino and Ricco (2021)	Greenbook forecasts and autocorrelation of surprises	Alternative identification, signalling channel	Closest analogue to the FIE construction
Bauer and Swanson (2023)	Surprises predictable from public news; the Fed responds to news	Benchmark identification	Provides decomposition and interpretation
Cieslak and Schrimpf (2019)	Non-monetary news in central bank communication	Related	Shows separation of components in FOMC announcements
Ilzetzki and Jin (2021)	Puzzling change in international transmission	Contradicted and reconciled	Data and specification replicated
Degasperi, Hong and Ricco (2020)	Global transmission of US monetary policy with information control	Similar methodology, different focus	Focuses on channels, not the information component
Miranda-Agrippino and Rey (2020)	Global financial cycle driven by US monetary policy	Related, shows effect on global risk assets	Frames why spillovers matter
Plagborg-Moller and Wolf (2021)	Local projections and VARs estimate the same responses	Methodological support	Justifies using both methods
Stock and Watson (2018)	Identification by external instruments	Methodological foundation	Framework for treating surprises as instruments
Pagan (1984)	Inference with generated regressors	Methodological critique	Applies to the two-step construction of the components

Source: reviewer's analysis; the connections to papers cited in the text come from the paper's own references, the others are added by the reviewer.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** Separating the FIE channel from the global-cycle channel. *Why it matters:* It determines whether the finding is about information or about omitted cycle variables. *Existing limitation:* Controls omit X_t. *Way forward:* Horse-race and placebo tests, rolling out-of-sample components.
- **Gap:** Size and units of the shock and of the responses. *Why it matters:* Economic significance cannot be judged. *Existing limitation:* Normalisation not stated in the text. *Way forward:* Report responses per standard deviation, and as elasticities.
- **Gap:** Formal comparison of the two responses. *Why it matters:* The central claim is a contrast. *Existing limitation:* Each response is tested against zero. *Way forward:* Test the difference with joint covariance and a bootstrap that includes the first stage.

10.2 Literature-level gaps

- **Gap:** Which interpretation of the information component is right. *Why it matters:* Policy lessons differ for signalling, disclosure and reaction to news. *Existing limitation:* Different schemes yield similar signs. *Way forward:* Use survey expectations and text data to discriminate.
- **Gap:** Time variation in spillovers. *Why it matters:* The claim of rising spillovers has policy implications. *Existing limitation:* Different samples have different composition. *Way forward:* Time-varying parameter local projections with a fixed country set.

10.3 Methodological gaps

- **Gap:** Inference with generated, common shocks. *Why it matters:* The panel contains one shock series. *Existing limitation:* Clustering by month only. *Way forward:* Two-stage bootstrap and time-series inference robust to serial correlation (Lag-augmented local projections).
- **Gap:** Identification from cross-sectional exposure. *Why it matters:* It uses variation not tied to the global cycle. *Existing limitation:* The paper never includes time fixed effects. *Way forward:* Interact the components with country exposure measures and absorb the common shock with month effects.

11 Research Opportunities

Idea 1. Horse race: information component against the global cycle

RESEARCH QUESTION. Does the response of foreign output and equities to r_t^{FIE} survive when X_t and the first-stage predictors enter directly as controls?

MOTIVATION. Addresses the single most important weakness of the paper.

HYPOTHESIS. The FIE coefficients on industrial production and equities shrink toward zero once the global cycle is controlled, while the MP coefficients are unchanged.

ECONOMIC MECHANISM. If FIE is a proxy of global conditions, controlling directly for those conditions removes the effect.

LITERATURE CONNECTION. Bauer and Swanson (2023); Nakamura and Steinsson (2018); Miranda-Agrippino and Rey (2020).

DATA. The Bauer-Swanson surprise series and predictors, IMF International Financial Statistics, OECD.

VARIABLES. Outcomes of the paper, the six predictors, global activity indices.

METHOD. Panel local projections with X_t and its lags, and placebo components built from unrelated predictors.

IDENTIFICATION STRATEGY. Conditional exogeneity of the MP residual; comparison of coefficients across specifications.

EXPECTED CONTRIBUTION. A direct test of information against cycle.

MAIN RISK. Collinearity: X_t and r_t^{FIE} are nearly collinear, so the horse race may have little power.

Idea 2. Exposure interactions with month fixed effects

RESEARCH QUESTION. Do countries with greater US trade, commodity or dollar-debt exposure respond more to the MP and FIE components, once the common shock is absorbed by month effects?

MOTIVATION. Uses cross-sectional variation, which is less exposed to the global-cycle confound and solves the effective sample size problem.

HYPOTHESIS. Higher dollar-debt exposure amplifies depreciation and equity falls after MP; commodity exposure amplifies the FIE response.

ECONOMIC MECHANISM. Balance sheet and trade channels deliver exposure-dependent transmission; an information shock operates through demand for exports.

LITERATURE CONNECTION. Rey (2013); Miranda-Agrippino and Rey (2020); Kalemli-Ozcan (2019).

DATA. BIS locational statistics, UN Comtrade, IMF direction of trade, the paper's panel.

VARIABLES. Outcomes, exposure measures Z_i, components.

METHOD. $y_{i,t+h} = \gamma_h^{MP}(i_t^{MP} Z_i) + \gamma_h^{FIE}(i_t^{FIE} Z_i) + \lambda_t + \Gamma_i + \square_{i,t}$.

IDENTIFICATION STRATEGY. Month fixed effects λ_t absorb the common shock; identification from differential exposure.

EXPECTED CONTRIBUTION. A mechanism-based test of the transmission channels.

MAIN RISK. Exposure may be correlated with other country characteristics; month effects remove the level effect that the paper estimates.

Idea 3. Real-time information component and announcement-day returns

RESEARCH QUESTION. Is the predictable part of the surprise, computed in real time, a priced and tradable signal for the dollar and foreign risk assets on FOMC days?

MOTIVATION. By construction i_t^{FIE} is a function of public data known before the meeting; if markets did not anticipate it, there is a predictable announcement return.

HYPOTHESIS. Real-time FIE predicts the sign of the announcement-day dollar and equity returns.

ECONOMIC MECHANISM. Markets under-react to the Fed reaction function, so the news-driven part of the surprise is unpriced until the announcement.

LITERATURE CONNECTION. Bauer and Swanson (2023); Kuttner (2001); Gurkaynak, Sack and Swanson (2005).

DATA. Daily exchange rates from the Federal Reserve H.10 release, equity indices, Eurodollar futures.

VARIABLES. Announcement-day returns, rolling out-of-sample FIE, MP residual.

METHOD. Event study with recursive estimation of Equation (1); then a simple long-short strategy.

IDENTIFICATION STRATEGY. Predictability test out of sample, avoiding look-ahead.

EXPECTED CONTRIBUTION. Links the macro decomposition to a tradable signal and tests market efficiency about the Fed reaction function.

MAIN RISK. About 250 meetings give low power; transaction costs and regime changes.

Idea 4. Expectations channel with survey forecasts

RESEARCH QUESTION. Do foreign growth and inflation forecasts rise after the FIE component and fall after the MP component?

MOTIVATION. Tests the claimed mechanism, which the paper does not test.

HYPOTHESIS. FIE raises forecasts of foreign GDP growth within the month, MP lowers them.

ECONOMIC MECHANISM. Belief revision about US and global demand.

LITERATURE CONNECTION. Nakamura and Steinsson (2018); Jarocinski and Karadi (2020); Miranda-Agrippino and Ricco (2021).

DATA. Consensus Economics monthly forecasts (licensed), IMF World Economic Outlook, central bank surveys.

VARIABLES. Forecast revisions of growth, inflation and exchange rate.

METHOD. Local projections of forecast revisions on both components.

IDENTIFICATION STRATEGY. Same as the paper, with revisions as outcomes, which are closer to the shock in time.

EXPECTED CONTRIBUTION. Separates information from cycle by looking at beliefs.

MAIN RISK. Data licensing and the short sample of surveys.

Idea 5. Inference correcting for generated shocks and testing antisymmetry

RESEARCH QUESTION. Is $\beta_b^{MP} + \beta_b^{FIE} = 0$ rejected, and are significance statements robust to first-stage estimation error?

MOTIVATION. The paper's central statement is a contrast, but it is not tested.

HYPOTHESIS. The two responses differ significantly in sign, but the antisymmetry hypothesis is rejected for some variables.

ECONOMIC MECHANISM. If antisymmetry fails, the Standard HFI is not a simple average of two mirror images.

LITERATURE CONNECTION. Pagan (1984); Plagborg-Moller and Wolf (2021); Stock and Watson (2018).

DATA. The paper's dataset or a public replication.

VARIABLES. Impulse responses with covariance matrices.

METHOD. Two-stage block bootstrap including the first stage; joint Wald test across horizons.

IDENTIFICATION STRATEGY. Unchanged.

EXPECTED CONTRIBUTION. Correct inference for a widely used decomposition.

MAIN RISK. Modest novelty.

11.1 Ranking

Table 6. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Horse race	3	5	5	4	4	3	24
2. Exposure interactions	4	4	4	5	4	5	26
3. Real-time FIE and returns	4	4	4	4	4	4	24
4. Expectations channel	5	2	2	4	4	4	21
5. Corrected inference	2	5	5	5	3	3	23

Source: reviewer's subjective scoring.

The strongest three are Idea 2, Idea 1 and Idea 3. Idea 2 is first because it changes the source of identifying variation, uses public data and tests the transmission channel directly. Idea 1 is the most direct answer to the main weakness and is cheap. Idea 3 turns the decomposition into an out-of-sample test with a trading interpretation, which also addresses look-ahead.

12 Learning Framework

12.1 Prerequisite concepts

- High-frequency identification of monetary shocks and the Fed Information Effect.
- Local projections and their relation to vector autoregressions.
- Bayesian vector autoregressions with the Minnesota prior.
- Open economy macroeconomics: Mundell-Fleming, Dornbusch overshooting, the dollar's role.
- Ordinary least squares projection algebra: fitted values, residuals, orthogonality.

12.2 Core concepts learned

- A policy surprise is a mixture of a policy shock and a reaction to news; their foreign effects can have opposite signs.
- Omitting one component yields a variance-weighted average of the two responses.
- Sign counts across countries give a distribution-free summary of heterogeneity.

12.3 Important equations

Equations (1), (2) and (3) for the decomposition; Equation (5) for the local projection; Equation (6) for the weighted average; Equation (10) for the Bayesian VAR with an exogenous shock.

12.4 Methods learned

Panel local projections with country-specific lag controls and time-clustered errors; Bayesian VARs with exogenous shocks; pooled and mean-group panel VAR; decomposition of a shock by projection on predetermined information.

12.5 Questions to test understanding

COMPREHENSION

1. What are i_t^{MP} and i_t^{FIE} and why are they orthogonal?
2. Why would the unadjusted surprise generate a positive industrial production response?
3. Which five variables define the sample and why were they chosen?

ANALYSIS

1. Derive Equation (6) and state when it fails.
2. Why does clustering by month matter when the shock is common to all countries?
3. What bias arises from using fitted values and residuals from a first stage as regressors?

RESEARCH

1. Design a test that separates information from the global cycle.
2. How would the conclusions change if w varies over time?
3. Which data would let you test the expectations channel directly?

12.6 Papers to read next

Bauer and Swanson (2023); Jarocinski and Karadi (2020); Miranda-Agrippino and Ricco (2021); Ilzetzki and Jin (2021); Plagborg-Moller and Wolf (2021).

13 Conclusion

13.1 One-sentence thesis

The international effect of a US rate surprise changes sign depending on whether the surprise is the residual or the news-predicted part, and the unadjusted surprise averages the two.

13.2 Three most important insights

- The Standard HFI coefficient is a variance-weighted average, so its sign depends on how much of the surprise is predictable from news.
- Opposite-sign responses are found across the large majority of countries, sub-samples and estimators.
- The information interpretation is not tested directly and may overlap with the global cycle.

13.3 Strongest evidence

The replication on the Ilzetzki and Jin sample and specification, together with the 90 to 97.5 percent country counts in the SVARs.

13.4 Biggest weakness

The FIE component is a fitted index of public global-cycle variables, estimated in sample and not controlled for directly.

13.5 Most important research gap

Which interpretation of the information component is correct, and whether the effect survives controlling for the cycle.

13.6 Best research opportunity

Exposure interactions with month fixed effects (Idea 2), complemented by a real-time test of the FIE component (Idea 3).

13.7 Three papers to read next

Bauer and Swanson (2023); Jarocinski and Karadi (2020); Ilzetzki and Jin (2021).

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